Linear Algebra

Lecture 08

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Important Theorems

Theorem 2.2.4

If E be an elementary matrix

(a) If E results from multiplying a row of I_n by k, then $\det(E) = k$.

(b) If E results from interchanging two rows of I_n, then $\det(E) = -1$.

(c) If E results from adding a multiple of one row of I_n to another, then $\det(E) = 1$.

Lemma 2.3.2

If B is $n \times n$ matrix and E is an elementary matrix of order $n \times n$, then $\det(EB) = \det(E)\det(B)$.

Proof: We pay attention to each type of elementary row operation which is available and divide the proof into different cases.
Case-1: If E is obtained by multiplying a row of I_n with a scalar of k, then by using theorem

> If the elementary matrix E results from performing a certain row operations on I_m and if A is an $m \times n$ matrix, then the product EA is the matrix that results when the same row operation is performed on A.

This means that

$$EB = C\,(say)$$
$$\det(EB) = \det(C)$$

Using the part(a) of last theorem above, we get

$$\det(EB) = k\det(B)$$

Since $\det(E) = k\det(I) = k$,

$$\det(EB) = \det(E)\det(B)$$

Case-2: We can prove by using the other type of E.R.O.S to get to the same conclusion. For that we have to make use of the last theorem part(b) and part(c). (Do it yourself)

Theorem 2.2.3

A square matrix A is invertible if and only if $\det(A) \neq 0$.

Case-1: Consider A is invertible and we would like to show that $\det(A) \neq 0$. Consider R is the row-reduced echelon form of A. As a preliminary step, we will show that $\det(A)$ and $\det(R)$ are both zero or both nonzero. Let $E_1, E_2, \cdots, E_r$ be the elementary matrices that correspond to the elementary operations that produce R from A. Thus

$$E_r E_{r-1} \cdots E_2 E_1 A = R$$

by using Lemma 2.3.2,

$$\det(R) = \det(E_r)\det(E_{r-1})\cdots\det(E_2)\det(E_1)\det(A)$$

By using Theorem 2.2.4, the determinant of elementary matrices are all nonzero. It means that $\det(A)$ and $\det(R)$ are both zero or nonzero. Now we move to main body of proof.

If A is invertible then by Fundamental Theorem of Algebra $\{(a) \implies (b) \implies (c)\}$, we have $R = I$, so $\det(R) = \det(I) = 1 \neq 0$ and consequently $\det(A) \neq 0$.

Case-2: Conversely, consider if $\det(A) \neq 0$ and we would like to show that A is invertible. Since $\det(A) \neq 0$, this means that $\det(R) \neq 0$, so R cannot have a row of zeros. It follows from Theorem

Theorem 1.4.3

If R is the reduced echelon form of an $n \times n$ matrix, then either R has a row zeros or R is the identity matrix.

This implies $R = I$, so A is invertible.

Theorem 2.3.4

If A and B are square matrices of same size, then

$$\det(AB) = \det(A)\det(B)$$

We divide the proof into two parts depending on whether A is invertible or not.

Case-1: If A is not invertible, then by using the Theorem

Theorem 1.6.5

Let A and B be square matrices of same size. If AB is invertible, then A and B must also be invertible.

This means that AB is not invertible. Using the previous theorem

$$\det(AB) = 0$$

and $\det(A) = 0$ This implies

$$\begin{aligned}\det(AB) &= 0 \cdot \det(B)\\ \det(AB) &= \det(A)\det(B)\end{aligned}$$

Case-2: If A is invertible then by using Fundamental Theorem of Algebra, A can be expressed as a product of elementary matrices such that

$$\begin{aligned}A &= E_r E_{r-1} \cdots E_2 E_1\\ AB &= E_r E_{r-1} \cdots E_2 E_1 B\\ \det(AB) &= \det(E_r E_{r-1} \cdots E_2 E_1 B)\\ \det(AB) &= \det(E_r)\det(E_{r-1}) \cdots \det(E_2)\det(E_1)\det(B)\\ \det(AB) &= \det(A)\det(B)\end{aligned}$$

Fundamental Theorem of Linear Algebra

Theorem 2.5.3 (Equivalent Statments)

If A is $n \times n$ matrix then the following statements are equivalent, that is, all true, or all false

(a) A is invertible.

(b) $Ax = 0$ has only the trivial solution.

(c) The reduced row-echelon form of A is I_n.

(d) A is expressible as a product of elementary matrices.

(e) $Ax = b$ is consistent for every $n \times 1$ matrix b.

(f) $Ax = b$ has exactly one solution for every $n \times 1$ matrix b.

(g) $\det(A) \neq 0$.